Pearson Edexcel International A Level · Pure Mathematics 3

# Chapter 8 — Numerical methods
# Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第8章 · 数值方法
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

**1.** P3-C08-R01 · P3 · Chapter 8

Let $f(x) = x^3 + 2x - 7$.

(a) Show that $f(x) = 0$ has exactly one real root, and that it lies between $1$ and $2$. (4)

(b) The iteration is $x_{n+1} = \sqrt[3]{7 - 2x_n}$, starting with $x_1 = 1.5$.

(i) Find $x_2, x_3, x_4$ to $6$ decimal places. (3)

(ii) Continue to obtain the root to $3$ decimal places and verify the rounding using a sign-change interval. (3)

(Total for Question 1 is 10 marks)

For $x > 0$, $f(x) = x + \ln x - 3$.

(a) Show that it has exactly one root in $(2, 3)$. (4)

(b) Use $x_{n+1} = 3 - \ln x_n$ with $x_1 = 2$ to find $x_2, x_3, x_4$, and then the root to 3 decimal places. (4)

(Total for Question 2 is 8 marks)

The curves $y = e^{-x}$ and $y = x$ meet at $x = \alpha$.

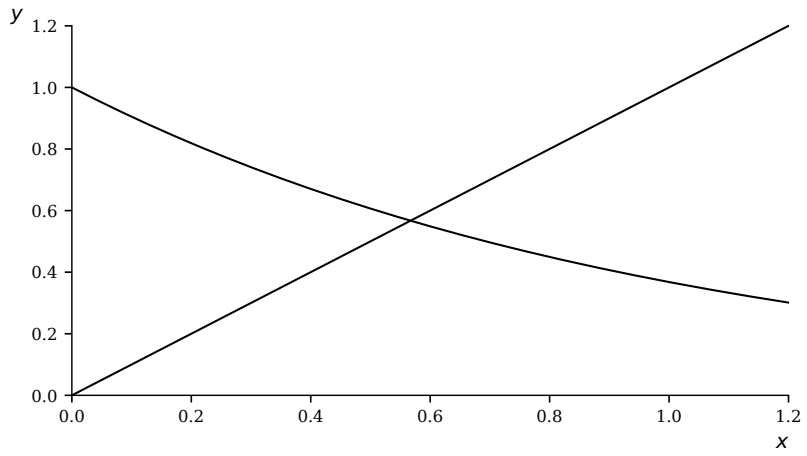



Figure 3

(a) Explain why there is exactly one intersection for real $x$ and locate it in $(0, 1)$. (4)

(b) Use $x_{n+1} = e^{-x_n}$, $x_1 = 0.5$, to calculate $x_2$ to $x_5$ to 6 decimal places and give $\alpha$ to 4 decimal places. (4)

(Total for Question 3 is 8 marks)

$f(x) = x\cos x - 0.4$ on $0 \le x \le 0.6$.

(a) Show that a root lies in $(0.4, 0.5)$. (3)

(b) Derive a fixed-point formula $x_{n+1} = 0.4/\cos x_n$. Starting with $x_1 = 0.45$, find $x_2, x_3, x_4$ to 6 decimal places. (4)

(Total for Question 4 is 7 marks)

A decaying response is $R(t) = (t+1)e^{-t}$ for $t \geq 0$. A detector switches off when $R = 0.2$.

(a) Show that the switching equation can be written $t = \ln(5(t+1))$ and explain why it has one positive solution. (4)

(b) Use this iteration from $t_1 = 3$ to find $t_2, t_3, t_4$ to 6 decimal places and the switching time to 3 decimal places. (4)

(Total for Question 5 is 8 marks)

$f(x) = (x^2 - 2)^2$.

(a) Explain why $f(1)f(2) > 0$ does not prove there is no root in $(1, 2)$. Find the root exactly. (3)

(b) Use $x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$ with $x_1 = 1$ to find $x_2, x_3, x_4$ to 6 decimal places. (3)

(Total for Question 6 is 6 marks)

$f(x) = 1/(x - 1)$.

(a) Calculate $f(0.9)$ and $f(1.1)$ and explain why their opposite signs do not establish a root between them. (3)

(b) Show algebraically that $f$ has no real zero. (3)

(Total for Question 7 is 6 marks)

The equation $x^2 - x - 2 = 0$ has a positive root $\alpha$.

(a) Find both exact roots and derive $x = \sqrt{x + 2}$ for the positive root. (3)

(b) Use initial value $x_1 = 1$.

(i) Find $x_2, x_3, x_4$ for $x_{n+1} = \sqrt{x_n + 2}$. (2)

(ii) Find $x_2, x_3$ for $x_{n+1} = x_n^2 - 2$ and explain why this does not approximate the requested root. (3)

(Total for Question 8 is 8 marks)

$f(x) = x^3 - 4x + 1$.

(a) Show that there is a root in each interval $(-3, -2)$, $(0, 1)$ and $(1, 2)$. (3)

(b) Explain why these account for every real root, and calculate each to $3$ decimal places. (3)

(Total for Question 9 is 6 marks)

On $x \geq 0$, the equation $\sin x = \ln(x+1)$ has the root $x = 0$ and another root $\alpha \in (1.6, 1.8)$.

(a) Verify the stated bracket for the non-zero root. (3)

(b) Use $x_{n+1} = e^{\sin x_n} - 1$ from $x_1 = 1.7$ to obtain $x_2, x_3, x_4$ to 6 decimal places. Explain why starting at 0 would not find $\alpha$. (4)

(Total for Question 10 is 7 marks)

$f(x) = x^3 + x - 5$. It is known to have a unique real root.

(a) Perform two bisections starting from $[1, 2]$, stating the retained interval after each. (4)

(b) Find the root to $3$ decimal places and verify it using the half-unit rounding boundaries. (3)

(Total for Question 11 is 7 marks)

The area beneath $y = 1 + e^x$ from 0 to $a > 0$ is 5.

(a) Show that $a = \ln(6 - a)$ and state the restriction needed for this iteration. (3)

(b) From $a_1 = 1.5$, find $a_2, a_3, a_4$ to 6 decimal places and the area-splitting input to 3 decimal places. (4)

(Total for Question 12 is 7 marks)

$f(x) = x^3 - 2x - 4$. A proposed root is $1.999$.

(a) Calculate the residual $f(1.999)$ and explain why a small residual does not mean that the decimal is an exact root. (3)

(b) Find the exact root in $[1, 3]$ and show that it is unique on this interval. (3)

(Total for Question 13 is 6 marks)

The iteration $x_{n+1} = \ln(4 - x_n)$ is used to solve $e^x + x - 4 = 0$.

(a) State the real-input restriction and explain why $x_1 = 5$ is invalid. (3)

(b) Using $x_1 = 1$, find $x_2, x_3, x_4$ to 6 decimal places and the root to 3 decimal places. (4)

(Total for Question 14 is 7 marks)

$f(x) = x^3 - 2x - 2$.

(a) Show that $f(x) = 0$ can be written $x = \sqrt[3]{2x + 2}$, and verify a root in $(1, 2)$. (3)

(b) Use the derived iteration with $x_1 = 1.5$ to find $x_2, x_3, x_4$ and then the root to 4 decimal places. (4)

(Total for Question 15 is 7 marks)

**16.** P3-C08-R16 · P3 · Chapter 8

An iteration is $x_{n+1} = (a + \ln x_n)/2$. Its fixed point is known to be $2$.

(a) Find $a$ exactly and the equation solved by this iteration. (3)

(b) For this $a$, start at $x_1 = 1.8$ and find $x_2, x_3, x_4$ to 6 decimal places. (3)

(Total for Question 16 is 6 marks)

$f(x) = (x^3 - 7)^3$.

(a) Find its exact real root and determine whether $f$ changes sign there. (3)

(b) Calculate $f(1.9125)$ and $f(1.9135)$, and use them to give the root correct to $3$ decimal places. (3)

(Total for Question 17 is 6 marks)

**18.**



A cost model is $C(q) = q + \ln q$ for positive integer output $q$. A project requires $C(q) \geq 7$.

(a) Show that the equation $C(x) = 7$ has one real solution for $x > 0$. (3)

(b) Approximate the real threshold to $3$ decimal places and find the least permitted integer output. (4)

(Total for Question 18 is 7 marks)

For $0 < x < \pi/2$, $\cot x = x$.

(a) Derive the iteration $x_{n+1} = \arctan(1/x_n)$ and verify a root in $(0.8, 0.9)$. (4)

(b) From $x_1 = 0.85$, find $x_2, x_3, x_4$ and then the root to $4$ decimal places. (4)

(Total for Question 19 is 8 marks)

$f(x) = 3x\cos x - 1$ on $(0, \pi/2)$.

(a) Show that there is one root in $(0.3, 0.4)$ and another in $(1.2, 1.4)$. (4)

(b) To approximate the larger root use $x_{n+1} = \arccos(1/(3x_n))$.

(i) state why $x_1 = 0.2$ is invalid, (2)

(ii) from $x_1 = 1.3$, find $x_2, x_3, x_4$ to 6 decimal places. (3)

(Total for Question 20 is 9 marks)

On $x \geq 0$, $\ln(2x+1) = x^2$ has the root 0 and a positive root $\alpha$.

(a) Verify that $\alpha \in (1, 1.2)$. (3)

(b) Use $x_{n+1} = \sqrt{\ln(2x_n + 1)}$, $x_1 = 1.1$, to find $x_2, x_3, x_4$ and $\alpha$ to 3 decimal places. (4)

(Total for Question 21 is 7 marks)

The iteration $x_{n+1} = \frac{1}{2}\cos x_n$ starts at $x_1 = 0.5$.

(a) Find $x_2$ to $x_6$, using 6 decimal places. (3)

(b) Let $f(x) = x - \frac{1}{2}\cos x$. Verify a sign change across the last two computed values and obtain the fixed point to 4 decimal places. (4)

(Total for Question 22 is 7 marks)

$f(x) = \ln x - (x-1)/2$, with $x > 0$. A calculator gives a proposed root $x = 3.5129$.

(a) Calculate $f(3.5129)$ to at least 6 significant figures and state whether this decimal is an exact root. (3)

(b) Find the other exact root and verify that a second root lies in $(3.5, 3.6)$. (4)

(Total for Question 23 is 7 marks)

A time $t$, measured in minutes, satisfies $t = \ln(t + 3)$. An iteration starts from $t_1 = 1.5$.

(a) Find $t_2, t_3, t_4$ to 6 decimal places. (3)

(b) If $s = 60t$ is measured in seconds, derive an equivalent iteration for $s$ and state the corresponding initial value. (3)

(Total for Question 24 is 6 marks)

The model equation is $x^3 + x = p$ for $x > 0$. The measured parameter satisfies $4.99 \le p \le 5.01$.

(a) Show that the positive solution increases as $p$ increases, using the monotonicity of $x^3 + x$. (3)

(b) Find the smallest and largest possible solutions to $4$ decimal places and determine whether the model fixes $x$ to $3$ decimal places. (4)

(Total for Question 25 is 7 marks)

The equation is $x = 1 + \sqrt{3 - x}$.

(a) Find the exact real solution, checking roots introduced by squaring. (4)

(b) Use $x_{n+1} = 1 + \sqrt{3 - x_n}$ with $x_1 = 1.5$ to find $x_2, x_3, x_4$ and their absolute errors relative to the exact root. (3)

(Total for Question 26 is 7 marks)

For $0 < x < \pi/2$, $\cos x = x/2$.

(a) Derive both fixed-point forms $x = 2\cos x$ and $x = \arccos(x/2)$. (3)

(b) Use $x_1 = 1$ in each iteration.

(i) Calculate $x_2, x_3, x_4$ for $x_{n+1} = 2\cos x_n$. (3)

(ii) Calculate $x_2, x_3, x_4$ for $x_{n+1} = \arccos(x_n/2)$ and find the root to $4$ decimal places. (4)

(Total for Question 27 is 10 marks)

$f(x) = \ln(x - 1) + x$ for $x > 1$.

(a) Find a sign-change interval of width $0.1$ containing a root, and prove this root is unique. (4)

(b) Use $x_{n+1} = 1 + e^{-x_n}$ from $x_1 = 1.25$ to obtain $x_2, x_3, x_4$ and the root to $4$ decimal places. (4)

(Total for Question 28 is 8 marks)

The function $f(x) = x + \ln x - 2$ is defined for $x > 0$.

(a) Show that $f(x) = 0$ has exactly one root in $1 < x < 2$. (3)

(b) The iteration $x_{n+1} = 2 - \ln x_n$ is used with $x_0 = 1.5$.

(i) Calculate $x_1, x_2, x_3$, giving each to $5$ decimal places. (2)

(ii) Show by a sign change that the root rounded to $2$ decimal places is $1.56$. (3)

(Total for Question 29 is 8 marks)

The equation $\cos x = x$ has a root $\alpha$ in $0 < x < 1$. Angles are in radians.

(a) Show that this root is unique. (3)

(b) Two rearrangements are proposed: $x_{n+1} = \cos x_n$ and $x_{n+1} = \arccos x_n$. Both start at $x_0 = 0.7$.

(i) Calculate the next two iterates for each rearrangement to $4$ decimal places. (2)

(ii) For the inverse-cosine iteration, the next iterate is approximately $0.8617$ and the following one is approximately $0.5321$. State why these four iterates do not justify reporting $\alpha$ to $3$ decimal places. (2)

(Total for Question 30 is 7 marks)

# Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

## Question 1

P3-C08-R01 · 10 marks

(a) · 4 marks

$f(1)=-4$, $f(2)=5$; $f'(x)=3x^2+2>0$.

多项式连续、端值异号给存在；导数恒正给至多一根。

| | |
|---|---|
| **M1** | 两异号值 |
| **M1** | 连续性 |
| **M1** | 导数恒正 |
| **A1** | 唯一根 |

(b) The iteration is $x_{n+1}=\sqrt[3]{7-2x_n}$, starting with $x_1=1.5$.

(i) · 3 marks

$1.587401, 1.563932, 1.570303$

逐次代入，不将中间值提前截为显示精度。

| | |
|---|---|
| **M1** | 正确首迭代 |
| **M1** | 顺序代入 |
| **A1** | 三值 |

(ii) · 3 marks

$1.569$; $f(1.5685)=-0.0041884559<0$, $f(1.5695)=0.0051968274>0$.

多次迭代近似根$1.568946\cdots$，用两个舍入边界代f所得异号，并结合(a)唯一性。

| | |
|---|---|
| **M1** | 迭代求近似 |
| **M1** | 舍入边界异号 |
| **A1** | 三位小数 |

## Question 2

P3-C08-R02 · 8 marks

(a) · 4 marks

$f(2) = \ln 2 - 1 < 0$, $f(3) = \ln 3 > 0$; $f' = 1 + 1/x > 0$.

域内连续、异号且严格递增。

| | |
|---|---|
| **M1** | 异号端值 |
| **M1** | 连续性 |
| **M1** | 正导数 |
| **A1** | 唯一性 |

(b) · 4 marks

$2.306853, 2.164116, 2.227988$; root $2.208$.

所有实际迭代输入保持正，逐项用未舍入值；继续到要求精度。

| | |
|---|---|
| **M1** | 正确公式代入 |
| **M1** | 三值 |
| **M1** | 持续迭代 |
| **A1** | 根 |

## Question 3

P3-C08-R03 · 8 marks

### (a) · 4 marks

$e^{-x} - x$ is strictly decreasing, with values 1 and $e^{-1} - 1$ at $0, 1$.

导数-e^{-x}-1恒负，多项函数连续且0、1异号。

| | |
|---|---|
| **M1** | 差函数 |
| **M1** | 负导数 |
| **M1** | 异号 |
| **A1** | 唯一性 |

### (b) · 4 marks

$0.606531, 0.545239, 0.579703, 0.560065$; $\alpha = 0.5671$.

先列四次迭代，给四位小数根时需继续计算，不能仅用前四次值。

| | |
|---|---|
| **M1** | 正确迭代 |
| **M1** | 四值 |
| **M1** | 继续计算 |
| **A1** | 根 |

## Question 4

P3-C08-R04 · 7 marks

(a) · 3 marks

$f(0.4) = -0.031575602,\ f(0.5) = 0.038791281$

两值异号，函数在该闭区间连续，故存在根。

| | |
|---|---|
| **M1** | 两端计算 |
| **M1** | 连续性 |
| **A1** | 存在根 |

(b) · 4 marks

$0.444224, 0.442995, 0.442737$

原式移项除cosx，在该区间cos正不为0；按式计算。

| | |
|---|---|
| **M1** | 合法等价重排 |
| **M1** | 正确首步 |
| **M1** | 后续计算 |
| **A1** | 三值 |

## Question 5

P3-C08-R05 · 8 marks

(a) · 4 marks

$e^{t}=5(t+1)$; $R'=-te^{-t}<0$ for $t>0$, with $R(0)=1$ and $R(4)=5e^{-4}<0.2$.

正量取ln合法；响应在正域严格下降，连续端值跨过门槛，所以唯一。

| | |
|---|---|
| **M1** | 等价对数重排 |
| **M1** | 单调导数 |
| **M1** | 阈值跨越 |
| **A1** | 唯一性 |

(b) · 4 marks

$2.995732, 2.994665, 2.994398$; $t=2.994$.

逐项迭代，继续至指定精度，时间域为非负。

| | |
|---|---|
| **M1** | 正确迭代 |
| **M1** | 三值 |
| **M1** | 继续 |
| **A1** | 时间 |

## Question 6

P3-C08-R06 · 6 marks

### (a) · 3 marks

$f(1)=1, f(2)=4$, but $f(\sqrt{2})=0$; the curve touches the axis without a sign change.

变号是存在根的充分检验，不变号不构成无根证明；本题偶重根只接触。

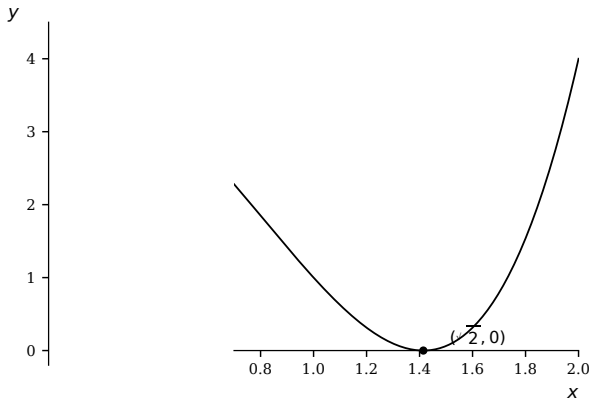
| | |
|---|---|
| **M1** | 端值与精确根 |
| **M1** | 接触不变号 |
| **A1** | 逻辑说明 |

### (b) · 3 marks

$1.500000, 1.416667, 1.414216$

给定的是固定点迭代；每一步输入均非零。

| | |
|---|---|
| **M1** | 正确首项 |
| **M1** | 递推值 |
| **A1** | 三值 |




Suggested sketch

## Question 7

P3-C08-R07 · 6 marks

(a) · 3 marks

$-10$ and 10; $f$ is not continuous on the interval because it is undefined at 1.

变号定理的连续前提失效，不能把渐近线穿越当作过轴。

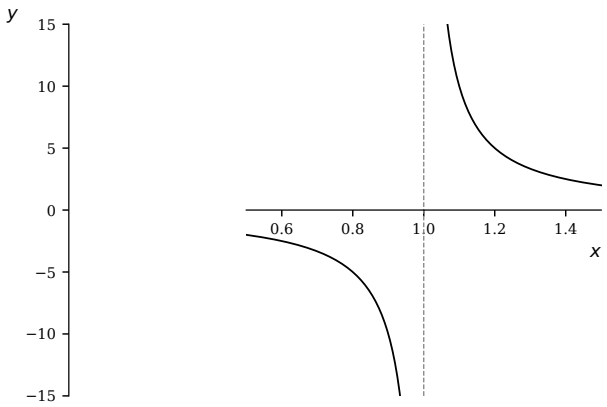
| | |
|---|---|
| **M1** | 两端值 |
| **M1** | 指出x1未定义 |
| **A1** | 连续前提说明 |

(b) · 3 marks

A non-zero numerator 1 divided by a finite non-zero denominator cannot be 0.

所有合法输入分子恒1，分式不可能为0。

| | |
|---|---|
| **M1** | 非零分子 |
| **M1** | 原域 |
| **A1** | 无根结论 |




Suggested sketch

## Question 8

P3-C08-R08 · 8 marks

(a) · 3 marks

Roots $2, -1$; positive branch gives $x = \sqrt{x+2}$.

原方程(x−2)(x+1)=0；平方根重排只保留非负分支。

| | |
|---|---|
| **M1** | 两根 |
| **M1** | 合法正分支 |
| **A1** | 迭代形式 |

(b) Use initial value $x_1 = 1$.

(i) · 2 marks

$1.732051, 1.931852, 1.982890$

逐步取正平方根。

| | |
|---|---|
| **M1** | 迭代 |
| **A1** | 三值 |

(ii) · 3 marks

$x_2 = -1, x_3 = -1$; it reaches the negative root instead.

等价固定点方程可有多个根，不同重排和起点可落入不同根；本题需求明确是正根。

| | |
|---|---|
| **M1** | 两步计算 |
| **M1** | 对照根符号 |
| **A1** | 失败原因 |

## Question 9

P3-C08-R09 · 6 marks

(a) · 3 marks

Values at $-3, -2, 0, 1, 2$ are $-14, 1, 1, -2, 1$, respectively.

三个不相交区间均连续且端值异号，分别至少一根。

| | |
|---|---|
| **M1** | 五个端值 |
| **M1** | 三次连续变号判断 |
| **A1** | 三根存在 |

(b) · 3 marks

Roots $-2.115,\ 0.254,\ 1.861$; a cubic has at most three distinct roots.

三个不同根已由互不相交区间保证，三次多项式不可能有第四个不同实根；用区间缩小或适当迭代计算。

| | |
|---|---|
| **M1** | 次数上界理由 |
| **M1** | 对三段数值求根 |
| **A1** | 三个近似 |

## Question 10

P3-C08-R10 · 7 marks

(a) · 3 marks

$f(1.6) = 0.044062158$, $f(1.8) = -0.055771786$

函数在1.6至1.8连续，两端差值异号。

| | |
|---|---|
| **M1** | 两差值 |
| **M1** | 连续性 |
| **A1** | 区间 |

(b) · 4 marks

$1.695719, 1.697182, 1.696687$; starting at $0$ stays at $0$.

对数方程指数还原得递推；0是该递推的另一固定点，起于0将永远不动。

| | |
|---|---|
| **M1** | 正确迭代 |
| **M1** | 三值 |
| **M1** | 检查零起点 |
| **A1** | 说明 |

## Question 11

P3-C08-R11 · 7 marks

(a) · 4 marks

$[1.5, 2]$, then $[1.5, 1.75]$.

f1.5=−.125、f1.75=2.109375；每次取异号的半区间。

| | |
|---|---|
| **M1** | 第一中点值 |
| **M1** | 第一保留区间 |
| **M1** | 第二中点及区间 |
| **A1** | 两区间 |

(b) · 3 marks

Root $1.516$; check $1.5155$ and $1.5165$.

缩小到同一舍入格内，并检查两格边界端值异号；唯一性已给。

| | |
|---|---|
| **M1** | 数值求根 |
| **M1** | 正确舍入格 |
| **A1** | 根 |

## Question 12

P3-C08-R12 · 7 marks

(a) · 3 marks

$a + e^a - 1 = 5 \Rightarrow e^a = 6 - a$; need $a < 6$.

先定积分得到面积方程，指数为正，取ln要求6−a正。

| | |
|---|---|
| **M1** | 正确积分 |
| **M1** | 等价重排 |
| **A1** | 对数限制 |

(b) · 4 marks

$1.504077, 1.503171, 1.503373$; $a = 1.503$.

逐步用未舍入值，解的语义是面积上界而非函数高度。

| | |
|---|---|
| **M1** | 正确公式 |
| **M1** | 三值 |
| **M1** | 继续计算 |
| **A1** | a |

## Question 13

P3-C08-R13 · 6 marks

(a) · 3 marks

$f(1.999) = -0.009994001$, $f(2) = 0$

确根要求f恰为0；1.999代入非零，只是近似；本题f2恰零。

| | |
|---|---|
| **M1** | 正确残差 |
| **M1** | 非零与确根区别 |
| **A1** | 说明 |

(b) · 3 marks

$x = 2$; $f' = 3x^2 - 2 > 0$ on $[1, 3]$.

验证f2零，区间导数正给最多一根。

| | |
|---|---|
| **M1** | 验证根 |
| **M1** | 正导数 |
| **A1** | 唯一性 |

## Question 14

P3-C08-R14 · 7 marks

(a) · 3 marks

$x_n < 4$ is required; at $5$ the logarithm has negative argument.

限制针对每一步送入递推的当前值，不能只检查最终根。

| | |
|---|---|
| **M1** | 真数正条件 |
| **M1** | 起值代入 |
| **A1** | 原因 |

(b) · 4 marks

$1.098612, 1.065189, 1.076643$; root $1.074$.

该起点的后续输入都小于$4$，逐次计算并继续。

| | |
|---|---|
| **M1** | 正确首步 |
| **M1** | 后续迭代 |
| **M1** | 根求值 |
| **A1** | 结果 |

## Question 15

P3-C08-R15 · 7 marks

(a) · 3 marks

$x^3 = 2x + 2$; $f(1) = -3$, $f(2) = 2$.

实立方根重排不增加或丢失实根，端值异号且连续。

| | |
|---|---|
| **M1** | 等价重排 |
| **M1** | 两端异号 |
| **A1** | 存在根 |

(b) · 4 marks

$1.709976, 1.756569, 1.766578$; root $1.7693$.

逐步迭代，四位小数根须继续而不是把x4直接当根。

| | |
|---|---|
| **M1** | 正确首步 |
| **M1** | 三值 |
| **M1** | 继续 |
| **A1** | 四位根 |

## Question 16

P3-C08-R16 · 6 marks

### (a) · 3 marks

$a = 4 - \ln 2$; $2x - \ln x - a = 0$ for $x > 0$.

固定点代2= (a+ln2)/2，反解参数；移项给原方程。

| | |
|---|---|
| **M1** | 固定点条件 |
| **M1** | 参数 |
| **A1** | 方程和域 |

### (b) · 3 marks

$1.947320, 1.986653, 1.996652$

参数保持精确4−ln2，不将它提前舍入。

| | |
|---|---|
| **M1** | 正确精确参数代入 |
| **M1** | 顺序迭代 |
| **A1** | 三值 |

## Question 17

P3-C08-R17 · 6 marks

(a) · 3 marks

$x = \sqrt[3]{7}$; yes, because both the inner cubic and its third power preserve sign.

三次外幂只在内部零点为0且保持正负号，所以穿越而非接触。

| | |
|---|---|
| **M1** | 精确零点 |
| **M1** | 奇次幂符号 |
| **A1** | 变号结论 |

(b) · 3 marks

$f(1.9125) = -1.0598645e-07$, $f(1.9135) = 2.4370558e-07$; root $1.913$.

两个数值虽很小但仍异号；这一区间是舍入到$1.913$的格，非仅按残差大小决定精度。

| | |
|---|---|
| **M1** | 边界准确代值 |
| **M1** | 异号与唯一根 |
| **A1** | 舍入值 |

## Question 18

P3-C08-R18 · 7 marks

(a) · 3 marks

$C' = 1 + 1/x > 0$; $C(5) < 7 < C(6)$.

正域连续严格递增，5和6跨门槛，故唯一临界值。

| | |
|---|---|
| **M1** | 正导数 |
| **M1** | 端值跨越 |
| **A1** | 唯一性 |

(b) · 4 marks

Threshold $5.327$; least $q = 6$.

可用x=7−lnx迭代，也可持续缩小区间；最终输出要满足整数模型而不是直接使用实根。

| | |
|---|---|
| **M1** | 正确数值方法 |
| **M1** | 根精度 |
| **M1** | 整数筛选 |
| **A1** | q |

## Question 19

P3-C08-R19 · 8 marks

(a) · 4 marks

$f(0.8) = 0.1712146,\ f(0.9) = -0.10644885$

正域内取倒比tanx=1/x，主值反正切落在原锐域，重排合法；cot差连续异号。

| | |
|---|---|
| **M1** | 合法反三角重排 |
| **M1** | 两端值 |
| **M1** | 连续域 |
| **A1** | 区间 |

(b) · 4 marks

$0.866302, 0.856914, 0.862302$; root $0.8603$.

按弧度计算，继续至要求精度。

| | |
|---|---|
| **M1** | 弧度迭代 |
| **M1** | 三值 |
| **M1** | 继续 |
| **A1** | 根 |

## Question 20

P3-C08-R20 · 9 marks

(a) · 4 marks

$f(0.3) = -0.14019716$, $f(0.4) = 0.10527319$; $f(1.2) = 0.30448792$, $f(1.4) = -0.286138$

两个不相交且连续的区间分别端值异号，至少各一根。

| | |
|---|---|
| **M1** | 第一对异号 |
| **M1** | 第二对异号 |
| **M1** | 连续性 |
| **A1** | 两根存在 |

(b) To approximate the larger root use $x_{n+1} = \arccos(1/(3x_n))$.

(i) · 2 marks

$1/(3x_1) = 5/3 > 1$, outside the domain of arccosine.

反余弦真数必须在$[-1,1]$，该起点超界。

| | |
|---|---|
| **M1** | 代起点检查真数 |
| **A1** | 域理由 |

(ii) · 3 marks

$1.311490, 1.313813, 1.314278$

每步用主值$\arccos$，迭代输入均为合法正数。

| | |
|---|---|
| **M1** | 正确首步 |
| **M1** | 顺序递推 |
| **A1** | 三值 |

## Question 21

P3-C08-R21 · 7 marks

(a) · 3 marks

$f(1)=0.098612289,\ f(1.2)=-0.21622457$

差函数连续，两个值异号。

| | |
|---|---|
| **M1** | 两端值 |
| **M1** | 连续性 |
| **A1** | 存在 |

(b) · 4 marks

$1.078495, 1.072203, 1.070339;\ \alpha=1.070.$

选择非负平方根与正根任务一致，起点不取另一个固定点$0$。

| | |
|---|---|
| **M1** | 正确正分支 |
| **M1** | 三值 |
| **M1** | 继续计算 |
| **A1** | 正根 |

## Question 22

P3-C08-R22 · 7 marks

(a) · 3 marks

$0.438791, 0.452633, 0.449649, 0.450300, 0.450158$

所有角用弧度，显示舍入但内部保留更多位。

| | |
|---|---|
| **M1** | 正确首步 |
| **M1** | 五次递推 |
| **A1** | 五值 |

(b) · 4 marks

$f(x_5) = 0.00014144376 > 0$, $f(x_6) = -3.0776166e-05 < 0$; root $0.4502$.

不能只因相邻迭代接近就说两者必为界；本题实际代入差函数核查符号后才构成变号区间。

| | |
|---|---|
| **M1** | 用未舍入相邻值检查 |
| **M1** | 连续变号 |
| **M1** | 继续至精度 |
| **A1** | 根 |

## Question 23

P3-C08-R23 · 7 marks

(a) · 3 marks

$f(3.5129) = -8.0928203e-06$, not zero; it is not an exact root.

直接代值检验，微小但非零的残差不能称确解。

| | |
|---|---|
| **M1** | 准确计算残差 |
| **M1** | 与零比较 |
| **A1** | 结论 |

(b) · 4 marks

$x = 1$; $f(3.5) = 0.0027629685$, $f(3.6) = -0.019066155$

原式代1为0；第二区间函数连续且端值异号，两个根不能混同。

| | |
|---|---|
| **B1** | 精确根1 |
| **M1** | 区间两端值 |
| **M1** | 连续性 |
| **A1** | 第二根存在 |

## Question 24

P3-C08-R24 · 6 marks

(a) · 3 marks

$1.504077, 1.504983, 1.505184$

以分钟作为递推变量，按给定公式计算。

| | |
|---|---|
| **M1** | 首步 |
| **M1** | 连续迭代 |
| **A1** | 三值 |

(b) · 3 marks

$s_{n+1} = 60\ln(s_n/60 + 3)$; $s_1 = 90$.

将t_n=s_n/60代入分钟公式，再两边乘60；不能只把ln外的单位改名。

| | |
|---|---|
| **M1** | 正确内部单位换算 |
| **M1** | 外部乘60 |
| **A1** | 公式和起值 |

## Question 25

P3-C08-R25 · 7 marks

(a) · 3 marks

Derivative $3x^2 + 1 > 0$, so larger output $p$ requires larger input $x$.

正域函数严格递增，一一对应，比较不同p对应x的顺序，无须引入隐函数求导。

| | |
|---|---|
| **M1** | 正导数 |
| **M1** | 一一单调对应 |
| **A1** | 顺序结论 |

(b) · 4 marks

Range approximately $[1.5147, 1.5172]$; no unique 3-decimal value.

分别数值求两个参数端值的根；它们舍入到三位小数不同，故测量不确定性已超过要求，更多迭代不能消除参数误差。

| | |
|---|---|
| **M1** | 两端参数求根 |
| **M1** | 根精度 |
| **M1** | 三位舍入比较 |
| **A1** | 结论 |

## Question 26

P3-C08-R26 · 7 marks

(a) · 4 marks

$x = 2$.

原式要求$1 \leqslant x \leqslant 3$；平方$(x-1)^2 = 3-x$得$(x-2)(x+1)=0$，$-1$不满足原式。

| | |
|---|---|
| **M1** | 原限制 |
| **M1** | 平方求两根 |
| **M1** | 排增根 |
| **A1** | 唯一根 |

(b) · 3 marks

$2.224745, 1.880486, 2.058071$; absolute errors $0.224745, 0.119514, 0.058071$.

分别将三个未舍入迭代值减2取绝对值，最终显示六位小数。

| | |
|---|---|
| **M1** | 正确三迭代 |
| **M1** | 绝对误差定义 |
| **A1** | 三误差 |

## Question 27

P3-C08-R27 · 10 marks

(a) · 3 marks

Both follow algebraically; the second uses the principal arccosine range containing the required root.

第一式乘2；第二式取主值反余弦，必须说明原根在其主值域。

| | |
|---|---|
| **M1** | 第一重排 |
| **M1** | 第二反三角重排 |
| **A1** | 分支说明 |

(b) Use $x_1 = 1$ in each iteration.

(i) · 3 marks

$1.080605, 0.941590, 1.177006$

分别递推，不因两式同解就假定迭代速度或行为相同。

| | |
|---|---|
| **M1** | 首步 |
| **M1** | 递推 |
| **A1** | 三值 |

(ii) · 4 marks

$1.047198, 1.019727, 1.035770$; root $1.0299$.

该序列继续计算可逼近所需根，使用弧度并保留内部精度。

| | |
|---|---|
| **M1** | 三值 |
| **M1** | 继续迭代 |
| **M1** | 原式核查 |
| **A1** | 根 |

## Question 28

P3-C08-R28 · 8 marks

(a) · 4 marks

$[1.2, 1.3]$; $f' = 1/(x-1) + 1 > 0$.

f1.2=ln.2+1.2<0，f1.3=ln.3+1.3>0，正域连续严格递增。

| | |
|---|---|
| **M1** | 两端异号 |
| **M1** | 连续性 |
| **M1** | 正导数 |
| **A1** | 唯一性 |

(b) · 4 marks

$1.286505, 1.276235, 1.279086$; root $1.2785$.

对数式指数还原得到迭代，后继值自动大于1，保持原定义域。

| | |
|---|---|
| **M1** | 合法重排 |
| **M1** | 三值 |
| **M1** | 继续及域 |
| **A1** | 根 |

## Question 29

P3-C08-R29 · 8 marks

(a) · 3 marks

Exactly one root.

$f(1) = -1 < 0$，$f(2) = \ln 2 > 0$，连续性给出一根；$f\prime(x) = 1 + 1/x > 0$说明严格递增，至多一根。

| | |
|---|---|
| **M1** | 两端异号及连续 |
| **M1** | 严格递增 |
| **A1** | 唯一性 |

(b) The iteration $x_{n+1} = 2 - \ln x_n$ is used with $x_0 = 1.5$.

(i) · 2 marks

$1.59453, 1.53342, 1.57250.$

逐次把未舍入前值代入$2 - \ln x_n$，只在报告数值时保留五位小数。

| | |
|---|---|
| **M1** | 正确迭代公式 |
| **A1** | 前三项 |

(ii) · 3 marks

$f(1.555) < 0 < f(1.565)$; root rounds to $1.56$.

计算$f(1.555) \approx -0.00352445$，$f(1.565) \approx 0.01288582$；唯一根位于舍入边界之间，所以保留两位小数为$1.56$。仅凭前几项迭代不能替代该精度检验。

| | |
|---|---|
| **M1** | 用舍入区间端点检验 |
| **A1** | 正确异号 |
| **B1** | 两位小数结论 |

## Question 30

P3-C08-R30 · 7 marks

(a) · 3 marks

Unique root in $(0,1)$.

令$f(x)=\cos x-x$，$f(0)=1>0$、$f(1)=\cos 1-1<0$；且$f\prime=-\sin x-1<0$，所以连续且严格递减，恰有一根。

| | |
|---|---|
| **M1** | 异号存在性 |
| **M1** | 导数符号 |
| **A1** | 唯一结论 |

(b) Two rearrangements are proposed: $x_{n+1}=\cos x_n$ and $x_{n+1}=\arccos x_n$. Both start at $x_0=0.7$.

(i) · 2 marks

Cosine: $0.7648, 0.7215$; inverse cosine: $0.7954, 0.6511$.

两种迭代都用弧度，分别连续代入两次。使用未舍入值计算第二步。

| | |
|---|---|
| **M1** | 正确执行两个不同迭代 |
| **A1** | 四个数值 |

(ii) · 2 marks

The iterates are not stable to three decimal places.

已列项仍有显著振荡，且差距远大于0.001，不能作为三位小数精度证据。题目未要求引用收敛判别定理。

| | |
|---|---|
| **B1** | 振荡或未稳定 |
| **B1** | 不足以证明三位精度 |